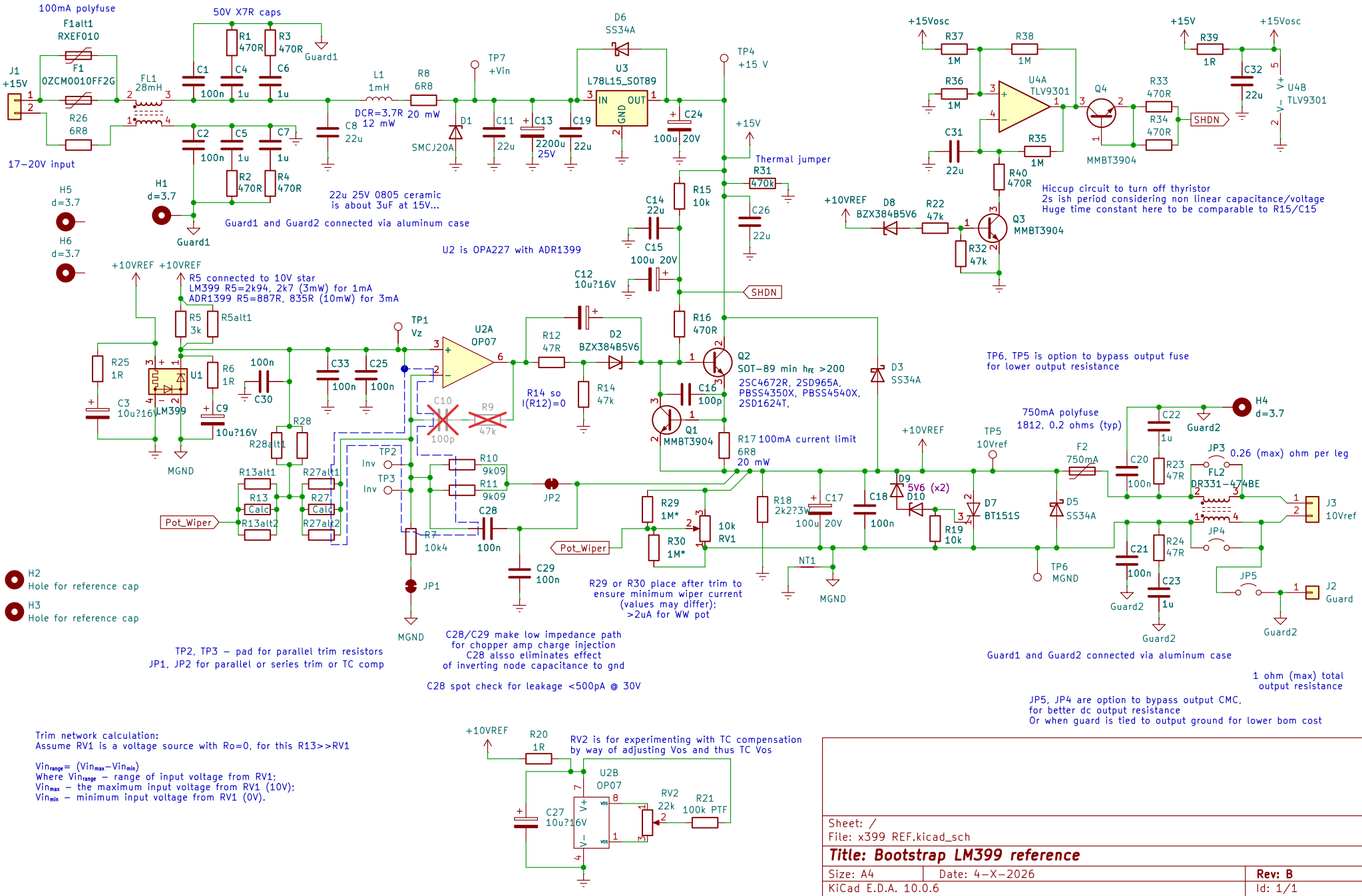

100mA polyfuse
F1alt1
RXEF010
F1
0ZCM0010FF2G
J1
+15V
17–20V input
R26
6R8
FL1
28mH
50V X7R caps
R1
470R
R3
470R
Guard1
C1
100n
C4
1u
C6
1u
C2
100n
C5
1u
C7
1u
R2
470R
R4
470R
H1
d=3.7
Guard1
H5
d=3.7
H6
d=3.7
C8
22u
L1
1mH
DCR=3.7R 12 mW
R8
6R8
20 mW
D1
SMCJ20A
TP7
+Vin
C11
22u
C13
2200u
25V
C19
22u
D6
SS34A
U3
L78L15_SOT89
IN
GND
OUT
C24
100u 20V
TP4
+15 V
+15V
Thermal jumper
R31
470k
C26
22u
22u 25V 0805 ceramic
is about 3uF at 15V...
Guard1 and Guard2 connected via aluminum case
U2 is OPA227 with ADR1399
C14
22u
C15
100u 20V
C12
10u?16V
R15
10k
R16
470R
SHDN
+10VREF +10VREF
R5 connected to 10V star
LM399 R5=2k94, 2k7 (3mW) for 1mA
ADR1399 R5=887R, 835R (10mW) for 3mA
R5
3k
R5alt1
R25
1R
U1
LM399
R6
1R
C3
10u?16V
C9
10u?16V
MGND
C30
100n
C33
100n
C25
100n
TP1
Vz
U2A
OP07
R12
47R
D2
BZX384B5V6
R14 so
I(R12)=0
R14
47k
C10
100p
R9
47k
Q2
SOT-89 min hFE >200
2SC4672R, 2SD965A,
PBSS4350X, PBSS4540X,
2SD1624T,
C16
100p
Q1
MMBT3904
R17 100mA current limit
6R8
20 mW
D3
SS34A
TP6, TP5 is option to bypass output fuse
for lower output resistance
R28
R28alt1
R13alt1
R13
Calc
R13alt2
R27alt1
R27
Calc
R27alt2
Pot_Wiper
TP2
Inv
TP3
Inv
R10
9k09
R11
9k09
JP2
C28
100n
R7
10k4
JP1
MGND
C29
100n
C28/C29 make low impedance path
for chopper amp charge injection
C28 alsso eliminates effect
of inverting node capacitance to gnd
C28 spot check for leakage <500pA @ 30V
TP2, TP3 – pad for parallel trim resistors
JP1, JP2 for parallel or series trim or TC comp
R29
1M*
R30
1M*
10k
RV1
Pot_Wiper
R29 or R30 place after trim to
ensure minimum wiper current
(values may differ):
>2uA for WW pot
R18
2k2?3W
C17
100u 20V
C18
100n
D9
5V6 (x2)
D10
R19
10k
D7
BT151S
+10VREF
TP5
10Vref
NT1
MGND
TP6
MGND
D5
SS34A
F2
750mA
750mA polyfuse
1812, 0.2 ohms (typ)
C22
1u
Guard2
H4
d=3.7
C20
100n
R23
47R
JP3
0.26 (max) ohm per leg
FL2
DR331-474BE
J3
10Vref
JP4
C21
100n
R24
47R
JP5
J2
Guard
Guard2
C23
1u
Guard2
Guard1 and Guard2 connected via aluminum case
1 ohm (max) total
output resistance
JP5, JP4 are option to bypass output CMC,
for better dc output resistance
Or when guard is tied to output ground for lower bom cost
H2
Hole for reference cap
H3
Hole for reference cap
+15Vosc
R37
1M
R38
1M
R36
1M
U4A
TLV9301
R35
1M
C31
22u
R40
470R
Q4
R33
470R
R34
470R
MMBT3904
SHDN
+15V
R39
1R
C32
22u
+15Vosc
U4B
TLV9301
Hiccup circuit to turn off thyristor
2s ish period considering non linear capacitance/voltage
Huge time constant here to be comparable to R15/C15
+10VREF
D8
BZX384B5V6
R22
47k
Q3
MMBT3904
R32
47k
Trim network calculation:
Assume RV1 is a voltage source with Ro=0, for this R13>>RV1
Vin_range = (Vin_max − Vin_min)
Where Vin_range – range of input voltage from RV1;
Vin_max – the maximum input voltage from RV1 (10V);
Vin_min – minimum input voltage from RV1 (0V).
+10VREF
R20
1R
U2B
OP07
C27
10u?16V
RV2
22k
R21
100k PTF
RV2 is for experimenting with TC compensation
by way of adjusting Vos and thus TC Vos
Sheet: /
File: x399 REF.kicad_sch
Title: Bootstrap LM399 reference
Size: A4
Date: 4–X–2026
Rev: B
KiCad E.D.A. 10.0.6
Id: 1/1